

ANTRIM COUNTY AP, MI, USA

WMO: 727436

Lat: 44.986N

Lon: 85.203W

Elev: 620

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 04863

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.3	(c) 2.8	(d) -9.4	(e) 3.4	(f) -0.5	(g) -3.7	(h) 4.6	(i) 5.2	(j) 17.1	(k) 25.4	(l) 15.8	(m) 24.5	(n) 0.5	(o) 20	(p) 0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	88.3	71.4	84.9	69.4	82.2	68.0	74.6	83.1	72.6	81.1	70.9	79.0	6.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	121.8	79.7	70.1	113.3	77.6	68.0	105.2	75.7	38.5	83.7	36.6	80.6	35.0	79.4	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	13.2	11.8	-13.5	93.3	7.7	2.1	-19.0	94.8	-23.5	96.0	-27.8	97.2	-33.4	98.7
			-13.9	77.2	7.6	2.5	-19.3	79.0	-23.7	80.4	-28.0	81.8	-33.5	83.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	22.2	22.9	30.5	42.7	54.6	63.5	68.0	66.6	60.4	48.6	38.1	28.3		
	DBStd	18.56	10.06	10.42	11.59	9.40	8.90	7.56	6.20	5.85	7.98	8.54	8.14	8.24		
	HDD50	3733	861	760	616	257	52	3	0	0	8	133	369	674		
	HDD65	7436	1326	1179	1070	669	344	115	41	50	182	513	808	1139		
	CDD50	2148	0	1	10	40	194	409	557	516	321	90	10	0		
	CDD65	380	0	0	0	2	21	72	134	101	45	5	0	0		
	CDH74	4271	0	0	14	40	317	841	1499	1041	459	59	1	0		
	CDH80	1184	0	0	2	7	86	249	440	274	116	10	0	0		
(14) Wind	WSAvg	4.3	5.0	4.8	4.6	5.1	4.5	3.6	3.2	3.2	3.6	4.4	5.0	5.1		
(15) Precipitation	PrecAvg	32.3	2.2	1.6	1.8	2.5	3.0	3.1	3.1	3.1	3.3	3.5	2.8	2.2		
	PrecMax	36.4	4.3	2.6	4.6	3.7	5.6	6.0	7.8	5.2	5.6	7.8	5.3	4.6		
	PrecMin	26.3	0.7	0.6	0.6	0.9	0.6	0.7	1.1	1.3	0.7	1.3	1.1	0.7		
	PrecStd	3.0	0.9	0.6	0.9	0.8	1.1	1.5	1.7	1.1	1.3	1.6	1.2	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.1	55.0	73.0	80.6	87.9	90.6	91.4	90.4	88.6	81.1	68.1	53.8		
		MCWB	44.5	50.0	59.3	60.4	68.2	72.8	72.9	72.5	70.5	66.0	56.9	50.8		
	2%	DB	43.5	45.8	61.1	72.3	81.8	86.4	88.1	86.3	82.5	73.4	60.8	46.2		
		MCWB	39.5	40.3	50.6	56.4	64.9	69.6	72.2	70.8	67.8	62.5	52.6	42.4		
	5%	DB	38.7	40.6	54.0	66.0	77.1	82.3	84.3	82.4	79.3	69.1	55.2	42.9		
		MCWB	36.0	36.5	46.9	52.1	61.4	67.5	70.2	68.7	66.1	60.6	48.7	39.2		
	10%	DB	36.1	36.6	47.6	61.0	72.2	79.2	81.8	80.4	75.1	63.5	51.7	39.1		
		MCWB	34.3	33.7	41.4	48.4	58.9	65.6	68.7	67.5	64.8	56.4	46.7	36.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.1	50.5	60.3	64.2	70.7	76.0	78.2	77.8	73.8	69.3	58.1	51.1		
		MCDB	46.8	54.1	71.1	74.0	82.4	86.3	86.9	85.4	82.8	77.9	65.0	53.5		
	2%	WB	39.7	40.7	53.3	58.5	67.7	72.4	75.0	73.5	71.0	65.4	54.1	43.9		
		MCDB	43.5	45.3	58.7	69.5	78.4	82.9	83.6	81.5	78.1	71.3	59.4	45.8		
	5%	WB	36.5	36.9	47.1	53.9	64.8	70.3	73.0	71.5	69.0	60.8	50.1	39.6		
		MCDB	38.5	40.3	52.3	63.2	73.0	79.4	81.1	79.4	76.1	67.1	55.2	42.5		
	10%	WB	34.0	33.9	41.5	49.6	61.2	67.9	71.0	69.5	66.6	57.3	46.4	36.6		
		MCDB	35.9	36.6	47.2	58.5	70.0	75.8	78.9	76.4	73.5	63.1	50.3	39.2		
(35) Mean Daily Temperature Range	MDBR		12.3	15.7	19.2	21.8	23.9	23.3	23.5	22.6	22.3	18.4	13.9	10.8		
	5% DB	MCDBR	14.5	20.0	26.7	32.4	30.7	27.7	26.7	26.2	27.0	26.6	22.4	15.6		
		MCWBR	12.3	15.4	18.0	18.1	15.2	13.4	12.7	13.0	13.8	16.4	15.8	13.5		
	5% WB	MCDBR	13.5	17.6	24.2	27.9	26.0	23.0	21.2	20.8	22.0	22.1	20.3	14.4		
		MCWBR	11.9	14.2	17.6	18.0	14.7	12.6	11.7	11.9	13.0	15.7	15.7	13.3		
(40) Clear-Sky Solar Irradiance	taub		0.287	0.295	0.320	0.352	0.383	0.398	0.393	0.386	0.374	0.340	0.304	0.280		
	taud		2.336	2.297	2.379	2.355	2.332	2.317	2.348	2.370	2.391	2.478	2.508	2.424		
	Ebn at Noon		262	282	290	288	281	276	276	274	268	264	257	254		
	Edh at Noon		24	31	33	37	39	40	38	36	32	26	21	21		
(44) All-Sky Solar Radiation	RadAvg		381	674	1127	1450	1763	1924	1938	1649	1280	753	413	283		
	RadStd		38	70	124	143	154	134	83	103	115	90	59	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page